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TREND-BASED MOCK PAPER

GATE Electronics and Communication Engineering

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Electronics and Communication Engineering
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Choose opposite of 'benevolent':

- (A) Hostile
- (B) Kind
- (C) Generous
- (D) Helpful

Q2. [1-MCQ] [1 mark]

If train 100m crosses pole in 10s, speed:

- (A) 10 m/s
- (B) 36 km/h
- (C) 18 km/h
- (D) 72 km/h

Q3. [1-MCQ] [1 mark]

Sum of first 100 natural numbers:

- (A) 1000
- (B) 5050
- (C) 4950
- (D) 5000

Q4. [1-MCQ] [1 mark]

Choose synonym of 'trivial':

- (A) Significant
- (B) Insignificant
- (C) Important
- (D) Major

Section B: Electronics and Communication Engineering

Q5. [2-NAT] [2 marks]

Mod-12 counter minimum FFs:

(Numerical Answer Type — Enter the numerical value)

Q6. [2-NAT] [2 marks]

4-bit ripple adder, 4 FAs at 20ns each. Delay:

(Numerical Answer Type — Enter the numerical value)

Q7. [2-MCQ] [2 marks]

$F(A,B,C,D) = \sum m(0,1,2,3,4,5,8,9,10)$. Minimal:

- (A) $A'+B'C$
 - (B) $A'D'+A'C'$
 - (C) $B'+A'D'$
 - (D) $A'B'+A'D'$
-

Q8. [2-NAT] [2 marks]

Setup time violation: solution:

(Numerical Answer Type — Enter the numerical value)

Q9. [2-MCQ] [2 marks]

2K x 8 ROM address lines:

- (A) 2
 - (B) 8
 - (C) 11
 - (D) 2048
-

Q10. [2-MCQ] [2 marks]

Half adder outputs:

- (A) Sum and carry
 - (B) Sum only
 - (C) Carry only
 - (D) Difference
-

Q11. [2-NAT] [2 marks]

4:1 MUX select lines:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-NAT] [2 marks]

$F(A,B,C,D) = \sum m(0,1,2,3,4,5,8,9,10)$. Minimal:

(Numerical Answer Type — Enter the numerical value)

Q13. [2-MCQ] [2 marks]

$Z_0=50\Omega$, $Z_L=100\Omega$. Reflection coefficient:

- (A) 0.33
- (B) 0.5
- (C) 0.67
- (D) 1.0

Q14. [2-NAT] [2 marks]

Skin depth in copper at 1MHz:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-MCQ] [2 marks]

Poynting vector $S =$

- (A) $E \times D$
 - (B) $E \times H$
 - (C) $H \times E$
 - (D) $B \times H$
-

Q16. [2-NAT] [2 marks]

Skin depth in copper at 1MHz:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-MCQ] [2 marks]

Intrinsic impedance of free space:

- (A) 120π ohm
 - (B) 377 ohm
 - (C) 50 ohm
 - (D) 75 ohm
-

Q18. [2-MCQ] [2 marks]

Poynting vector $S =$

- (A) $E \times D$
 - (B) $E \times H$
 - (C) $H \times E$
 - (D) $B \times H$
-

Q19. [2-NAT] [2 marks]

AM signal carrier 1kW, modulation 0.5. Total power:

(Numerical Answer Type — Enter the numerical value)

Q20. [2-MSQ] [2 marks]

Nyquist rate for $f_{\max}=4\text{kHz}$:

- (A) 4kHz
 - (B) 8kHz
 - (C) 2kHz
 - (D) 16kHz
-

Q21. [2-MCQ] [2 marks]

NBFM bandwidth approx:

- (A) $2f_m$
 - (B) $2(\Delta f + f_m)$
 - (C) Δf
 - (D) f_m only
-

Q22. [2-MCQ] [2 marks]

Shannon capacity: $BW=1\text{MHz}$, $SNR=15$. Capacity:

- (A) 4 Mbps

- (B) 1 Mbps
- (C) 15 Mbps
- (D) 5 Mbps

Q23. [2-NAT] [2 marks]

AM signal carrier 1kW, modulation 0.5. Total power:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-NAT] [2 marks]

AM signal carrier 1kW, modulation 0.5. Total power:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-MCQ] [2 marks]

Delta-Star transformation: R_a (star) =

- (A) $R_1 R_2 / (R_1 + R_2 + R_3)$
- (B) $(R_1 R_2 + R_2 R_3 + R_3 R_1) / R_3$
- (C) $R_1 + R_2$
- (D) $R_1 R_2 / R_3$

Q26. [2-MCQ] [2 marks]

Y-parameter Y_{11} for series impedance Z is:

- (A) $1/Z$
- (B) Z
- (C) $-Z$
- (D) 0

Q27. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q28. [2-MCQ] [2 marks]

Y-parameter Y_{11} for series impedance Z is:

- (A) $1/Z$
- (B) Z
- (C) $-Z$
- (D) 0

Q29. [1-MCQ] [1 mark]

$i^4 =$:

- (A) 0
- (B) 1
- (C) -1
- (D) i

Q30. [1-MCQ] [1 mark]

Integral of $1/x$ dx:

- (A) x
- (B) $\log(x) + C$

- (C) $1/x^2$
- (D) $x^{2/2}$

Q31. [1-MCQ] [1 mark]

$P(A \cap B) = ?$

- (A) $P(A) + P(B)$
- (B) $P(A) + P(B) - P(A \cap B)$
- (C) $P(A)P(B)$
- (D) $P(A)/P(B)$

Q32. [2-NAT] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q33. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q34. [1-MCQ] [1 mark]

Characteristic impedance $Z_0 =$:

- (A) $\sqrt{L/C}$
- (B) L/C
- (C) C/L
- (D) $\sqrt{LC}$

Q35. [2-MCQ] [2 marks]

Waveguide supports:

- (A) TEM
- (B) TE/TM
- (C) Both
- (D) None

Q36. [2-MCQ] [2 marks]

Skin depth increases with:

- (A) Frequency
- (B) Conductivity
- (C) Both
- (D) Neither

Q37. [2-MSQ] [2 marks]

Maxwell has how many equations:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q38. [2-NAT] [2 marks]

Parseval relates:

(Numerical Answer Type — Enter the numerical value)

Q39. [2-MCQ] [2 marks]

ROC of $X(z)=1/(1-0.5z^{-1})$ includes $z=2$. Signal:

- (A) Causal
 - (B) Anti-causal
 - (C) Two-sided
 - (D) Cannot determine
-

Q40. [2-MSQ] [2 marks]

Energy signal has:

- (A) Finite E, zero P
 - (B) Infinite E
 - (C) Finite P
 - (D) Both finite
-

Q41. [2-MCQ] [2 marks]

Input offset voltage causes:

- (A) Output saturates
 - (B) Output DC offset
 - (C) No effect
 - (D) BW increases
-

Q42. [2-NAT] [2 marks]

CE amp $h_{fe}=100$, $h_{ie}=1k$, $R_C=2k$. Voltage gain:

(Numerical Answer Type — Enter the numerical value)

Q43. [2-NAT] [2 marks]

CC (emitter follower) has:

(Numerical Answer Type — Enter the numerical value)

Q44. [1-MCQ] [1 mark]

RC oscillator frequency:

- (A) $f=1/(2RC)$
 - (B) $f=1/RC$
 - (C) $f=1/(4RC)$
 - (D) $f=2/RC$
-

Q45. [1-MCQ] [1 mark]

Ideal op-amp open-loop gain:

- (A) 0
 - (B) 1
 - (C) Infinite
 - (D) -1
-

Q46. [1-MCQ] [1 mark]

;

- (A) ; ∂f^3

- (B) ; $\partial f / \partial \pm$
- (C) I_c/I_b
- (D) I_e/I_c

Q47. [1-MCQ] [1 mark]

Common emitter has:

- (A) High input Z
- (B) Low input Z
- (C) Zero gain
- (D) Unity

Q48. [1-MCQ] [1 mark]

FET is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q49. [1-MCQ] [1 mark]

Zener diode used for:

- (A) Rectification
- (B) Voltage regulation
- (C) Amplification
- (D) None

Q50. [1-NAT] [1 mark]

SCR has:

(Numerical Answer Type — Enter the numerical value)

Q51. [1-MCQ] [1 mark]

MOSFET is:

- (A) BJT
- (B) Unipolar
- (C) Bipolar
- (D) None

Q52. [1-NAT] [1 mark]

n-type majority carriers:

(Numerical Answer Type — Enter the numerical value)

Q53. [1-MCQ] [1 mark]

ROM is:

- (A) Read only
- (B) Read write
- (C) Volatile
- (D) Cache

Q54. [1-MCQ] [1 mark]

4-bit binary counter counts:

- (A) 0-15
- (B) 0-9

- (C) 1-16
- (D) 1-10

Q55. [1-MCQ] [1 mark]

D flip-flop follows input after:

- (A) Same instant
- (B) One clock
- (C) Two clocks
- (D) Never

Q56. [1-NAT] [1 mark]

TTL stands for:

(Numerical Answer Type — Enter the numerical value)

Q57. [1-MCQ] [1 mark]

Routh-Hurwitz determines:

- (A) Stability
- (B) Time response
- (C) Frequency
- (D) Sensitivity

Q58. [1-MCQ] [1 mark]

Transfer function is defined:

- (A) In time domain
- (B) Laplace
- (C) Fourier
- (D) Z-domain

Q59. [1-MCQ] [1 mark]

Lead compensator:

- (A) Improves steady state
- (B) Improves transient
- (C) Both
- (D) None

Q60. [1-MCQ] [1 mark]

Nyquist criterion uses:

- (A) Polar plot
- (B) Root locus
- (C) Routh
- (D) Bode

Q61. [1-MCQ] [1 mark]

PCM stands for:

- (A) Pulse Code Modulation
- (B) Phase Code Modulation
- (C) Pulse Carrier Modulation
- (D) None

Q62. [1-MCQ] [1 mark]

AM stands for:

- (A) Amplitude Modulation
- (B) Angle Modulation
- (C) Analog Modulation
- (D) None

Q63. [1-NAT] [1 mark]

FM has:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

ASK modulation varies:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q65. [1-NAT] [1 mark]

Fourier of delta function:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	A	1
2	1MCQ	B	1
3	1MCQ	B	1
4	1MCQ	B	1
5	1MCQ	B	1
6	1MCQ	B	1
7	1MCQ	B	1
8	1NAT	B	1
9	1MCQ	B	1
10	1NAT	B	1
11	1MCQ	B	1
12	1NAT	B	1
13	1MCQ	A	1

14	1MCQ	C	1
15	1MCQ	C	1
16	1MCQ	B	1
17	1MCQ	B	1
18	1MCQ	A	1
19	1MCQ	A	1
20	1MCQ	B	1
21	1NAT	A	1
22	1MCQ	A	1
23	1MCQ	B	1
24	1MCQ	B	1
25	1MCQ	A	1
26	1MCQ	A	1
27	1MCQ	A	1
28	1NAT	A	1

29	1MCQ	A	1
30	1MCQ	A	1
31	2MCQ	B	2
32	2MCQ	D	2
33	2MSQ	C	2
34	2NAT	B	2
35	2NAT	A	2
36	2MCQ	A	2
37	2NAT	C	2
38	2NAT	A	2
39	2MSQ	B	2
40	2MCQ	B	2
41	2NAT	A	2
42	2MCQ	B	2
43	2NAT	A	2

44	2MCQ	D	2
45	2NAT	A	2
46	2MCQ	A	2
47	2MCQ	B	2
48	2MCQ	B	2
49	2MCQ	A	2
50	2MCQ	C	2
51	2MCQ	A	2
52	2NAT	A	2
53	2MCQ	A	2
54	2MCQ	A	2
55	2NAT	A	2
56	2NAT	A	2
57	2MCQ	A	2
58	2MSQ	A	2

59	2NAT	B	2
60	2NAT	A	2
61	2NAT	B	2
62	2NAT	B	2
63	2MCQ	B	2
64	2MCQ	D	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Complex Analysis | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: easy

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Probability | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Network, Signals & Systems | Topic: Fourier Transform | Difficulty: easy

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: Diodes | Difficulty: easy

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: SCR | Difficulty: moderate

Q11.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: MOSFET | Difficulty: difficult

Q12.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: Semiconductors | Difficulty: moderate

Q13.

Answer: A

Solution:

Correct Answer: A

Subject: Analog Circuits | Topic: Oscillators | Difficulty: easy

Q14.

Answer: C

Solution:

Correct Answer: C

Subject: Analog Circuits | Topic: Op-Amps | Difficulty: difficult

Q15.

Answer: C

Solution:

Correct Answer: C

Subject: Analog Circuits | Topic: BJT | Difficulty: difficult

Q16.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Circuits | Topic: Amplifiers | Difficulty: moderate

Q17.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Circuits | Topic: FET | Difficulty: moderate

Q18.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Memory | Difficulty: moderate

Q19.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Counters | Difficulty: moderate

Q20.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Circuits | Topic: Flip-Flops | Difficulty: moderate

Q21.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Logic Families | Difficulty: moderate

Q22.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Stability | Difficulty: easy

Q23.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Transfer Function | Difficulty: difficult

Q24.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Controllers | Difficulty: easy

Q25.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Stability | Difficulty: moderate

Q26.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Digital Comm | Difficulty: easy

Q27.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Modulation | Difficulty: easy

Q28.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Analog | Difficulty: moderate

Q29.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Digital | Difficulty: moderate

Q30.

Answer: A

Solution:

Correct Answer: A

Subject: Electromagnetics | Topic: Transmission Line | Difficulty: difficult

Q31.

Answer: B

Solution:

Correct Answer: B

Subject: Electromagnetics | Topic: Waveguides | Difficulty: easy

Q32.

Answer: D

Solution:

Correct Answer: D

Subject: Electromagnetics | Topic: Wave Propagation | Difficulty: easy

Q33.

Answer: C

Solution:

Correct Answer: C

Subject: Electromagnetics | Topic: Maxwell Equations | Difficulty: easy

Q34.

Answer: B

Solution:

$2^n \geq 12$. $n=4$ gives 16 states. Minimum 4 FFs.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: easy

Q35.

Answer: A

Solution:

Parseval: energy in time domain = energy in frequency domain.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: easy

Q36.

Answer: A

Solution:

$\Gamma = (100-50)/(100+50) = 50/150 = 1/3 = 0.33$.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q37.

Answer: C

Solution:

Carry ripples through 4 FAs. $t_{pd} = 4 \times 20 = 80\text{ns}$.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q38.

Answer: A

Solution:

$P_{\text{total}} = P_c(1+m^2/2) = 1000 \times (1+0.125) = 1125\text{W} = 1.125\text{kW}$.

Subject: Communication | Topic: Communication | Difficulty: easy

Q39.

Answer: B

Solution:

$f_s \geq 2 \times 4\text{kHz} = 8\text{kHz}$.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q40.

Answer: B

Solution:

$R_a = (R_1 R_2 + R_2 R_3 + R_3 R_1) / R_3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: difficult

Q41.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: B

Solution:

K-map: $F = A'(D + B'C) = A'D + A'B'C$. Closest: $A'D + A'C$.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q43.

Answer: A

Solution:

Setup: data not stable before clock. Lower clock freq gives longer period.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q44.

Answer: D

Solution:

Series Z: $I_1=I_2$, $V_1-V_2=Z \cdot I_1$. $Y_{11} = I_1/V_1$ with $I_2=0$ impossible ($I_1=I_2$). $Y_{11}=0$ for pure series.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q45.

Answer: A

Solution:

$\delta = \sqrt{2/(\omega \mu \sigma)}$. At 1MHz, δ approx 66 microns.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q46.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: B

Solution:

$S = E \times H$. Direction: power flow. Magnitude: power per unit area.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q48.

Answer: B

Solution:

Input offset: differential DC input causing DC output offset.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q49.

Answer: A

Solution:

NBFM: BW approx $2f_m$ (same as AM). WBFM: Carson rule.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q50.

Answer: C

Solution:

$2K = 2048$ locations. $\log_2 = 11$ address lines.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q51.

Answer: A

Solution:

Pole at $z=0.5$. $|z|>0.5$: ROC is $|z|>0.5$, causal.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q52.

Answer: A

Solution:

$A_v = -h_{fe} R_C / h_{ie} = -100 \cdot 2000 / 1000 = -200$.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q53.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Network Theory | Topic: Network Theory | Difficulty: difficult

Q54.

Answer: A

Solution:

$C = B \cdot \log_2(1+SNR) = 1e6 \cdot \log_2(16) = 1e6 \cdot 4 = 4$ Mbps.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q55.

Answer: A

Solution:

$$P_{\text{total}} = P_c(1+m^2/2) = 1000*(1+0.125) = 1125W = 1.125kW.$$

Subject: Communication | Topic: Communication | Difficulty: moderate

Q56.

Answer: A

Solution:

$\delta = \sqrt{2/(\omega \mu \sigma)}$. At 1MHz, δ approx 66 microns.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q57.

Answer: A

Solution:

Half adder: Sum=A XOR B, Carry=A AND B.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: easy

Q58.

Answer: A

Solution:

Energy: finite E, P=0. Power: finite P, E=infinity.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q59.

Answer: B

Solution:

4:1: $\log_2(4) = 2$ select lines.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q60.

Answer: A

Solution:

$$P_{\text{total}} = P_c(1+m^2/2) = 1000*(1+0.125) = 1125W = 1.125kW.$$

Subject: Communication | Topic: Communication | Difficulty: difficult

Q61.

Answer: B

Solution:

CC: $A_v \sim 1$, high input Z, high current gain, low output Z.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: difficult

Q62.

Answer: B

Solution:

K-map: $F = A'(D' + B'C') = A'D' + A'B'C'$. Closest: $A'D' + A'C'$.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q63.

Answer: B

Solution:

$$\eta_0 = \sqrt{\mu_0/\epsilon_0} = 377 \text{ ohm approx } 120\pi.$$

Subject: EMFT | Topic: EMFT | Difficulty: difficult

Q64.

Answer: D

Solution:

Series Z: $I_1=I_2$, $V_1-V_2=Z*I_1$. $Y_{11} = I_1/V_1$ with $I_2=0$ impossible ($I_1=I_2$). $Y_{11}=0$ for pure series.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q65.

Answer: B

Solution:

$S = E \times H$. Direction: power flow. Magnitude: power per unit area.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Digital Electronics: 8 questions, 16 marks
EMFT: 6 questions, 12 marks
Communication: 6 questions, 12 marks
Network Theory: 4 questions, 8 marks
Engineering Mathematics: 5 questions, 7 marks
Electromagnetics: 4 questions, 7 marks
Signal Systems: 3 questions, 6 marks
Analog Electronics: 3 questions, 6 marks
Analog Circuits: 5 questions, 5 marks
General Aptitude: 4 questions, 4 marks
Electronic Devices: 4 questions, 4 marks
Digital Circuits: 4 questions, 4 marks
Control Systems: 4 questions, 4 marks
Communication Systems: 4 questions, 4 marks
Network, Signals & Systems: 1 questions, 1 marks

Difficulty Distribution:

Easy: 16 | Moderate: 39 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.